Voicing and Airstream

Dr. Josef Fruehwald

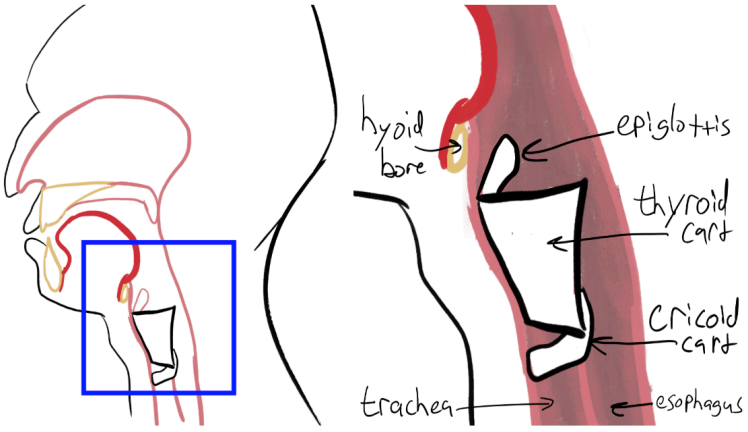
2026-09-08

Voicing

Voicing

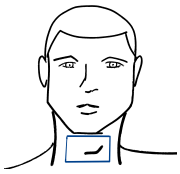
Previously:

The Larynx

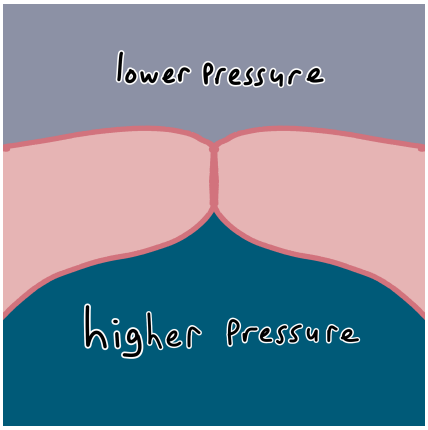


Voicing

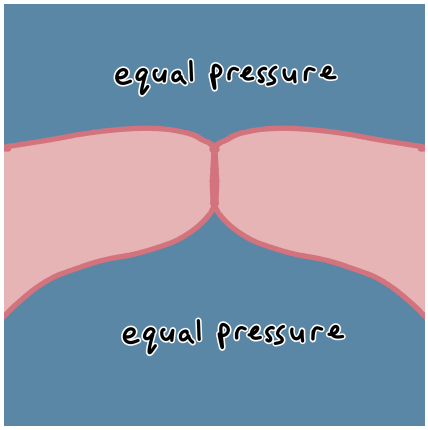
Voicing requires a pressure differential.



voicing possible

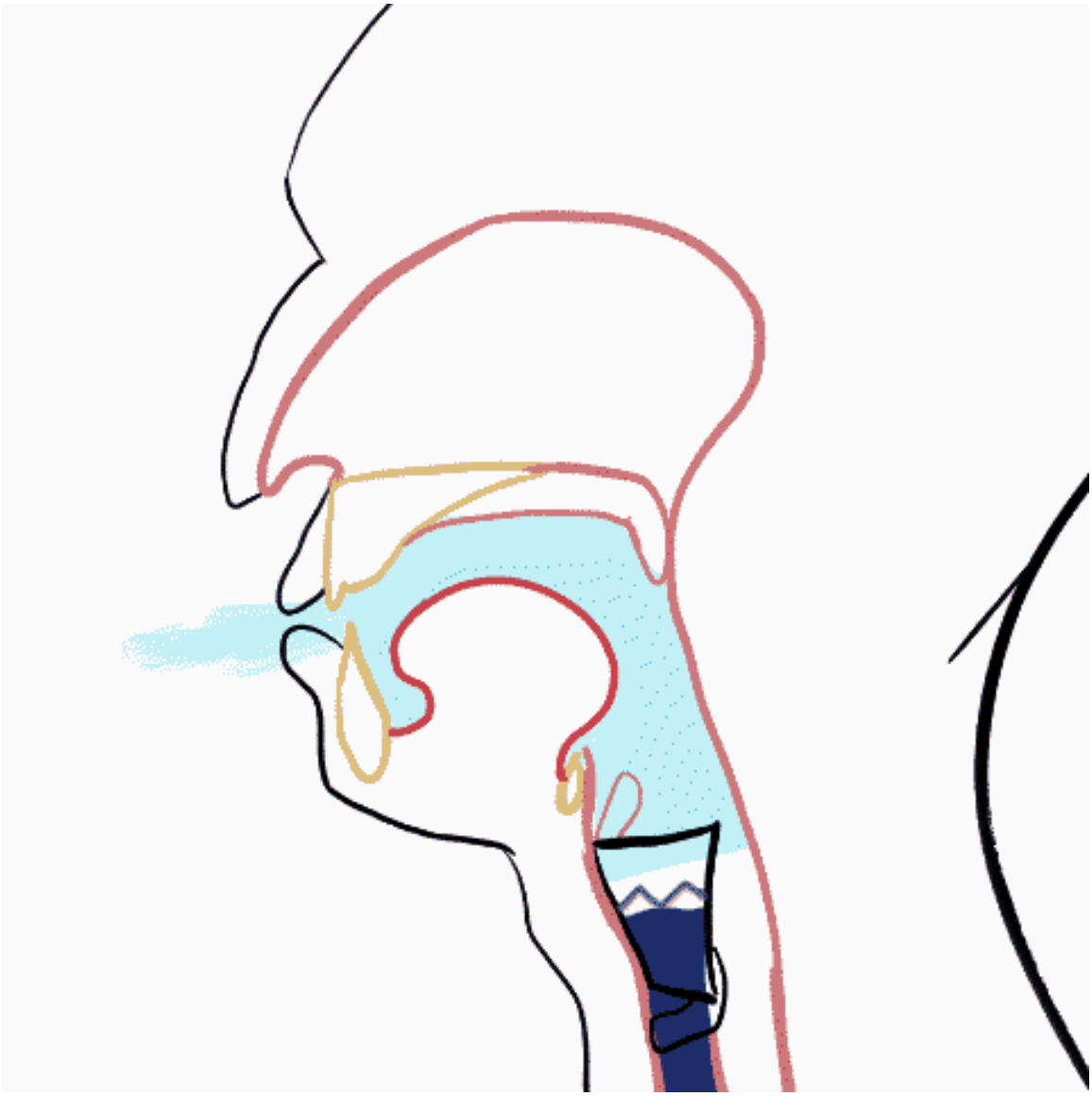


voicing not possible



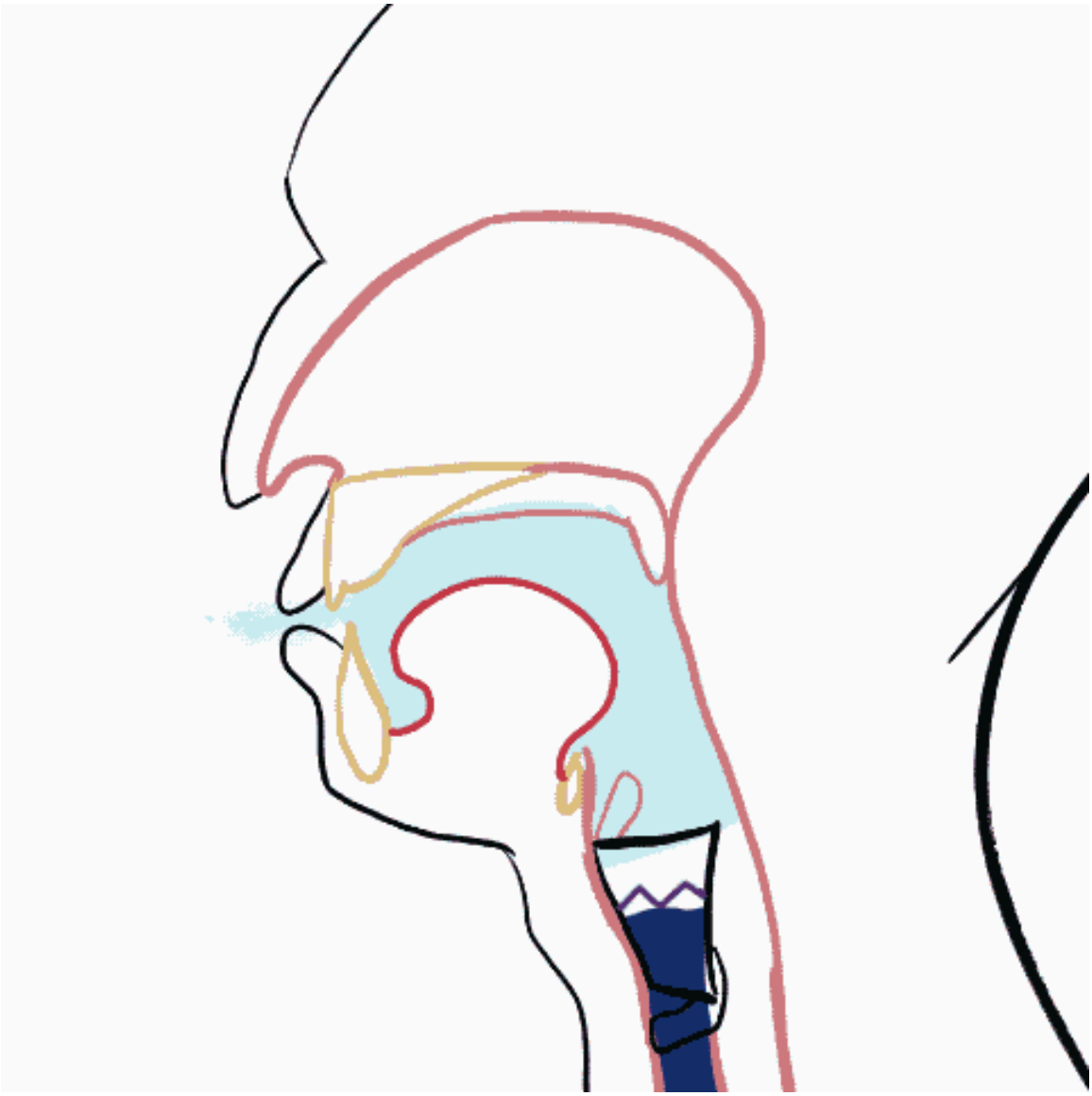
Voicing

No problem for sounds with incomplete closure.



Voicing

With complete closure, air pressure equalizes.

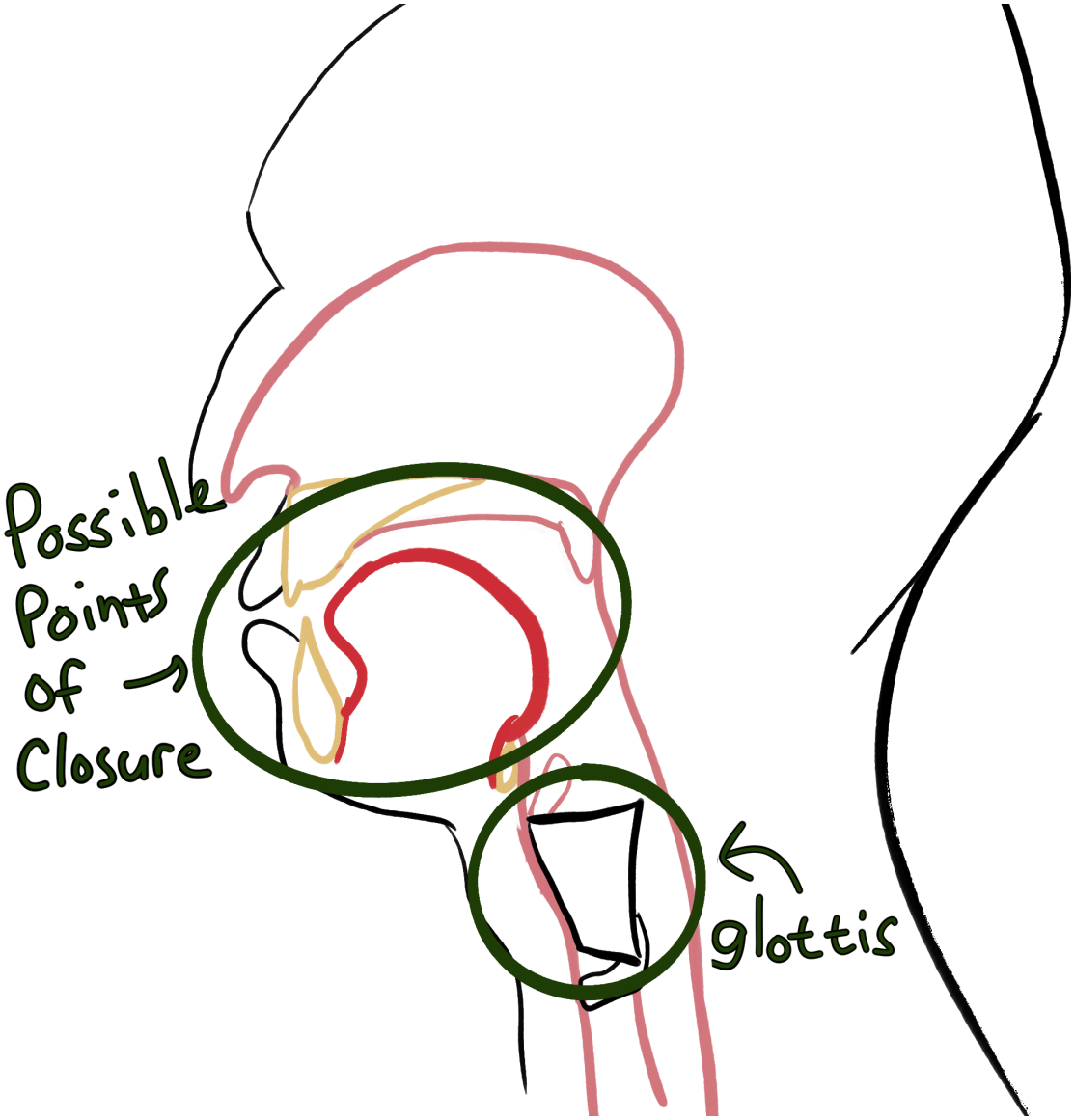


Voice Onset Time

Pulmonic Stops

Pulmonic	Egressive	Stop
“Air pressure from lungs”	“outward airflow”	“full closure”

Pulmonic Stops



Pulmonic Stops

The articulators involved with making stops don't *spatially* interfere with the glottis.

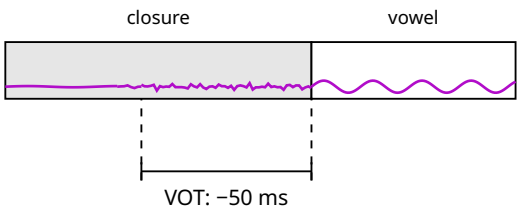
How might constrictions and the glottis interact *temporally*?

Fully Voiced Stops



Fully Voiced Stops

Also known as “negative voice onset time”

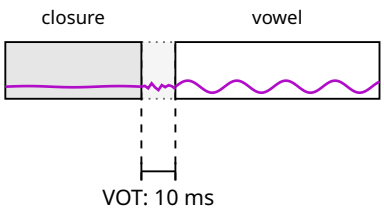


Short Lag Voice Onset Time



Short Lag Voice Onset Time

There is a brief lag between releasing the closure and the onset of voicing.

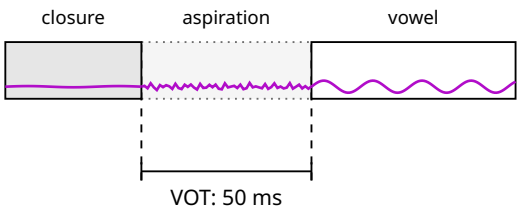


Long Lag Voice Onset Time



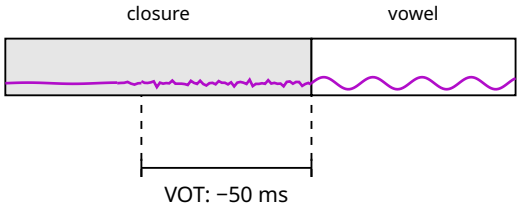
Short Lag Voice Onset Time

There is a long lag between releasing the closure and the onset of voicing.

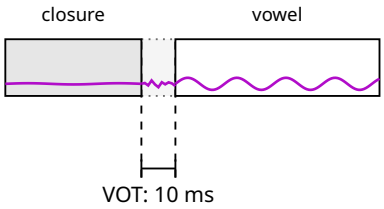


Voice Onset Time

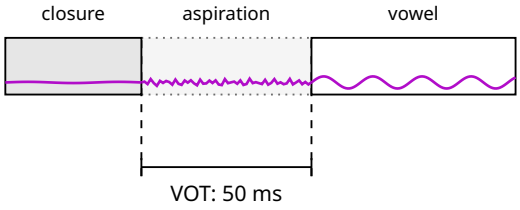
Fully voiced stops



Short lag (a.k.a. Unaspirated Stops)



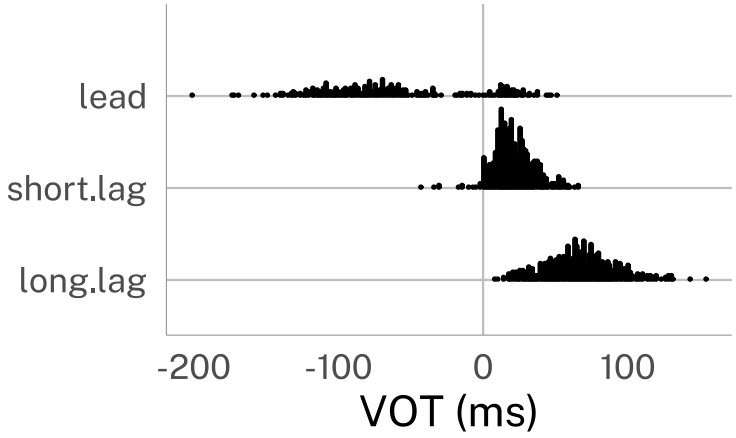
Long lag (a.k.a. Aspirated Stops)



Voice Onset Time

```
vot |>
  filter(
    #poa.broad == "coronal"
  ) |>
  mutate(
    vot.category = fct_reorder(vot.category, desc(vot))
  ) |>
  ggplot(
    aes(vot, vot.category)
  ) +
  geom_vline(
    xintercept = 0,
    linewidth = 0.4,
    color = "grey"
  ) +
  stat_dots(
    color = "black",
```

```
    fill = "black"
  ) +
  labs(
    y = NULL,
    x = "VOT (ms)"
  ) +
  theme_sub_panel(
    grid.major.y = element_line(
      linewidth = 0.4,
      color = "grey"
    )
  )
```



Chodroff et al. (2019)

Transcribing different VOTs

VOT	IPA
fully voiced	[b]
short lag	?
long lag	[pʰ]

Transcribing VOTs

	fully voiced	short lag	long lag
English, German		[b d g]	[pʰ tʰ kʰ]
Cantonese		[p t k]	[pʰ tʰ kʰ]

	fully voiced	short lag	long lag
Spanish, Dutch, Hungarian	[b d g]	[p t k]	
Eastern Armenian, Thai	[b d g]	[p t k]	[pʰ tʰ kʰ]

Lisker & Abramson (1964)

Transcribing VOT

Lesson: A lot of transcription is analysis in disguise!

Non-Pulmonic Stops

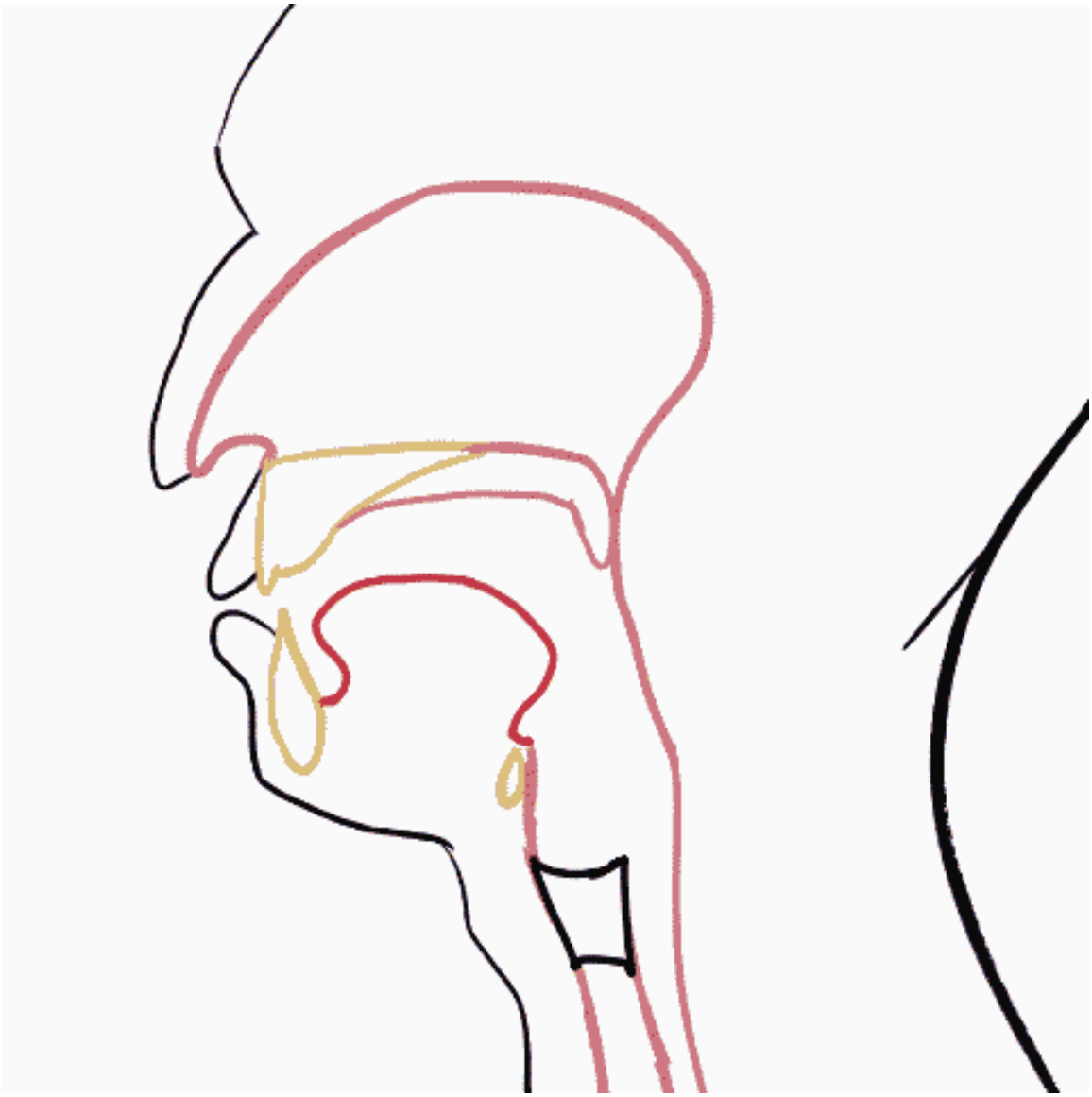
Non-Pulmonic Stops

	egressive (+ air pressure)	ingressive (-air pressure)
glottalic (closure at the glottis)	“ejectives”	“implosives”
velaric (closure at the velum)		“clicks”

Ejectives

1. An oral closure is made.
2. The glottis is tightly closed.
3. The larynx is raised.
4. Pressure builds behind the oral closure.
5. The oral closure is released.
6. The built up pressure rushes out.

Ejectives



Ejectives

You can have an ejective at any point of closure, and it's transcribed with [‘] (a regular apostrophe)

bilabial	alveolar	retroflex	palatal	velar	uvular
[pʼ]	[tʼ]	[ʈʼ]	[cʼ]	[kʼ]	[qʼ]

Ejective fricatives and affricates are also possible.

Practicing Ejectives

Try holding your breath (thus closing your glottis) and articulating “t t t t” repeatedly.

Implosives

1. An oral closure is made.
2. The glottis is tightly closed.
3. The larynx lowers.
4. Pressure drops behind the oral closure.
5. The oral closure is released.
6. Air rushes in, normalizing the low pressure.

Implosives



Implosives

Implosives can be made at any point of closure, and their IPA symbols tend to have a little hooktop.

bilabial	alveolar	velar	uvular
[ɓ]	[ɗ]	[ɠ]	[ʛ]

Clicks

1. A closure is made somewhere in front of the velum.
2. While that closure is maintained, the tongue body moves back.
3. A vacuum is created between the point of closure and the velum.
4. The closure is released.
5. Air rushes in normalizing air pressure.

Clicks



Clicks

You can only have clicks with active articulators that don’t interfere with the tongue body.

bilabial	dental	palatal	lateral	post-alveolar
[ʘ]	[ǀ]	[ǂ]	[ǁ]	[ǃ]

Clicks In Xhosa [ǁoːsa]

https://www.youtube.com/watch?v=n-4U2hfMpnk



Bibliography

Chodroff, E., Golden, A., & Wilson, C. (2019). Covariation of stop voice onset time across languages: Evidence for a universal constraint on phonetic realization. *The Journal of the Acoustical Society of America*, *145*(1), EL109–EL115. https://doi.org/10.1121/1.5088035

Lisker, L., & Abramson, A. S. (1964). A Cross-Language Study of Voicing in Initial Stops: Acoustical Measurements. *WORD*, *20*(3), 384–422. https://doi.org/10.1080/00437956.1964.11659830